

7(a)(i)	(part of) the output is combined with the input	M1
	reference to potential/voltage/signal	A1
7(a)(ii)	- increased (operating) stability - increased bandwidth/range of frequencies over which gain is constant - less distortion (of output) <i>Any 2 points.</i>	B2
7(b)(i)	1. gain = $3.6 / (48 \times 10^{-3})$	C1
	= 75	A1
	2. gain = $1 + R_F / R$ $75 = 1 + (92.5 \times 10^3) / R$	C1
	$R = 1300 \Omega$	A1
7(b)(ii)	for 68 mV, gain $\times V_{IN} = 5.1$ (V) or output voltage would be greater than the supply voltage	M1
	amplifier would saturate (at 5.0 V) or output voltage = 5.0 (V)	A1